

NC Hotel Group | F26 DIIG

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A booking-time risk lens for a reversible pilot

The decision

Which reservations create the largest avoidable cancellation/no-show exposure—and what should NC

20,000 reservation rows | Jul 2015–Aug 2017 | 2 properties

Audit before action

37.69% cancellation/no-show label rate

3,486 exact duplicate rows beyond first (1,338 groups)

Status and label are perfectly consistent

Duplicates are not silently deleted: they change the answer.

Headline: a material sensitivity

Raw rows: 37.69% label rate | \$2.815m gross exposure proxy

Exact-deduplicated: 31.00% | \$2.219m

ADR × stay nights is a gross exposure proxy—not lost revenue.

Where risk sits

City Hotel: 42.73% vs Resort: 27.91%

Groups: 63.05% cancellation/no-show label rate

City × Groups: 72.09%

Property mix and segment composition matter.

Lead time is the clearest gradient

0 days: 6.70%

31–90 days: 37.86%

181+ days: 58.45%

Long-lead bookings are a natural early-confirmation experiment population—not a reason to reject de

Signals, with causal caution

Non Refund: 99.49% (n=2,554)

Returning guest: 14.85% vs 38.47% non-returning

These are associations. Deposit/channel changes may alter conversion, mix, fees, and service.

Can we target risk?

Chronological logistic model

Test AUC 0.733 | Brier 0.197 vs 0.239 baseline

Frozen Non Refund rule: 100% test label rate; 18.6% of test labels

Monitor calibration drift; verify history timing.

Limitation: what this file cannot answer

No room count or inventory/capacity

No commissions, replacement demand, walk costs, or fee recovery

No causal policy assignment or experiment

Therefore: no safe overbooking multiplier, profit lift, or channel-shift claim.

Recommendation | targeted pilot

1. Audit duplicate identity and field timing
2. Randomize confirmation/deposit offer for eligible long-lead/high-risk bookings
3. Block by property, channel, lead band, arrival month
4. Measure conversion, kept nights, refunds, contacts, net contribution
5. Add inventory + economics before capacity decisions